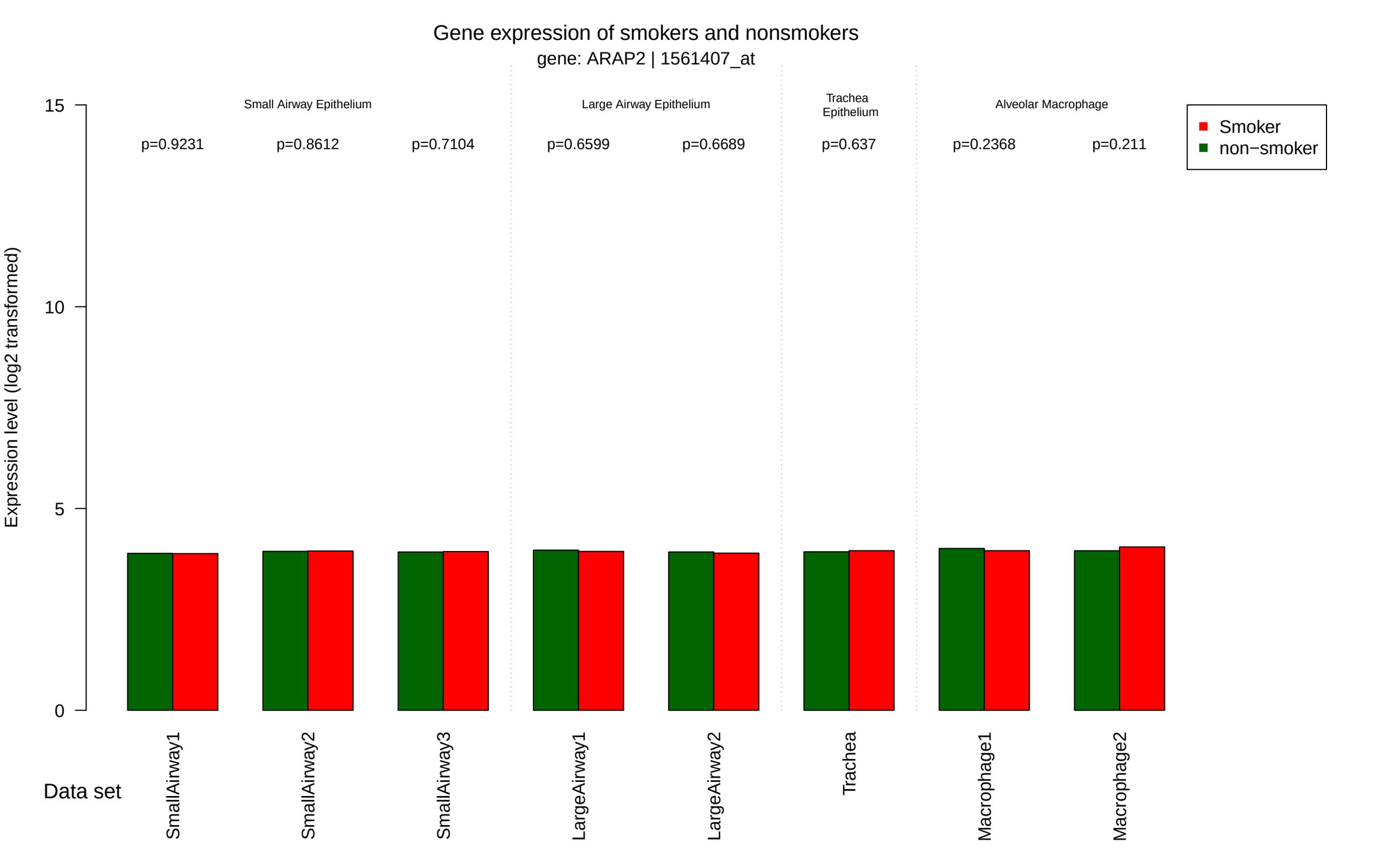
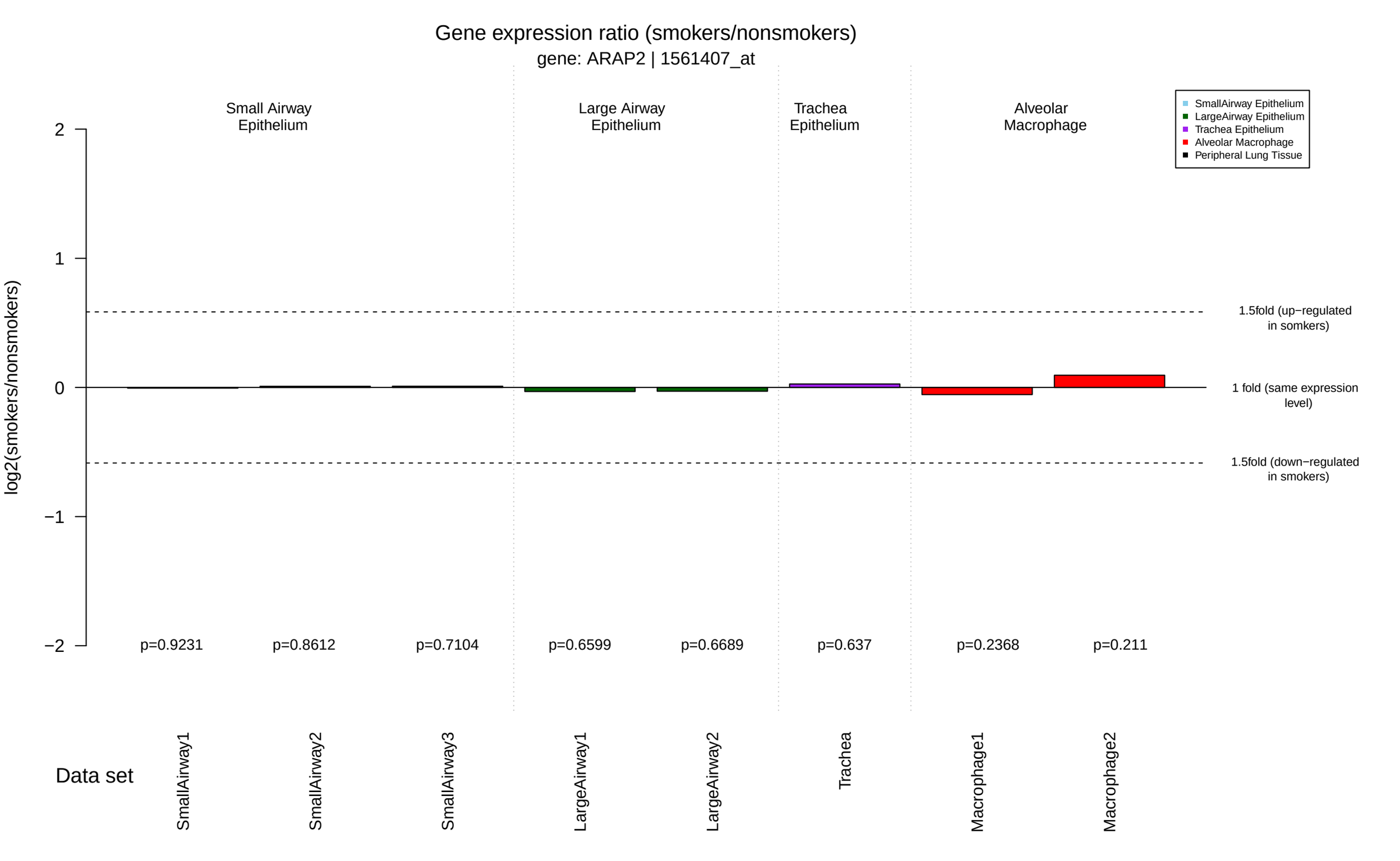
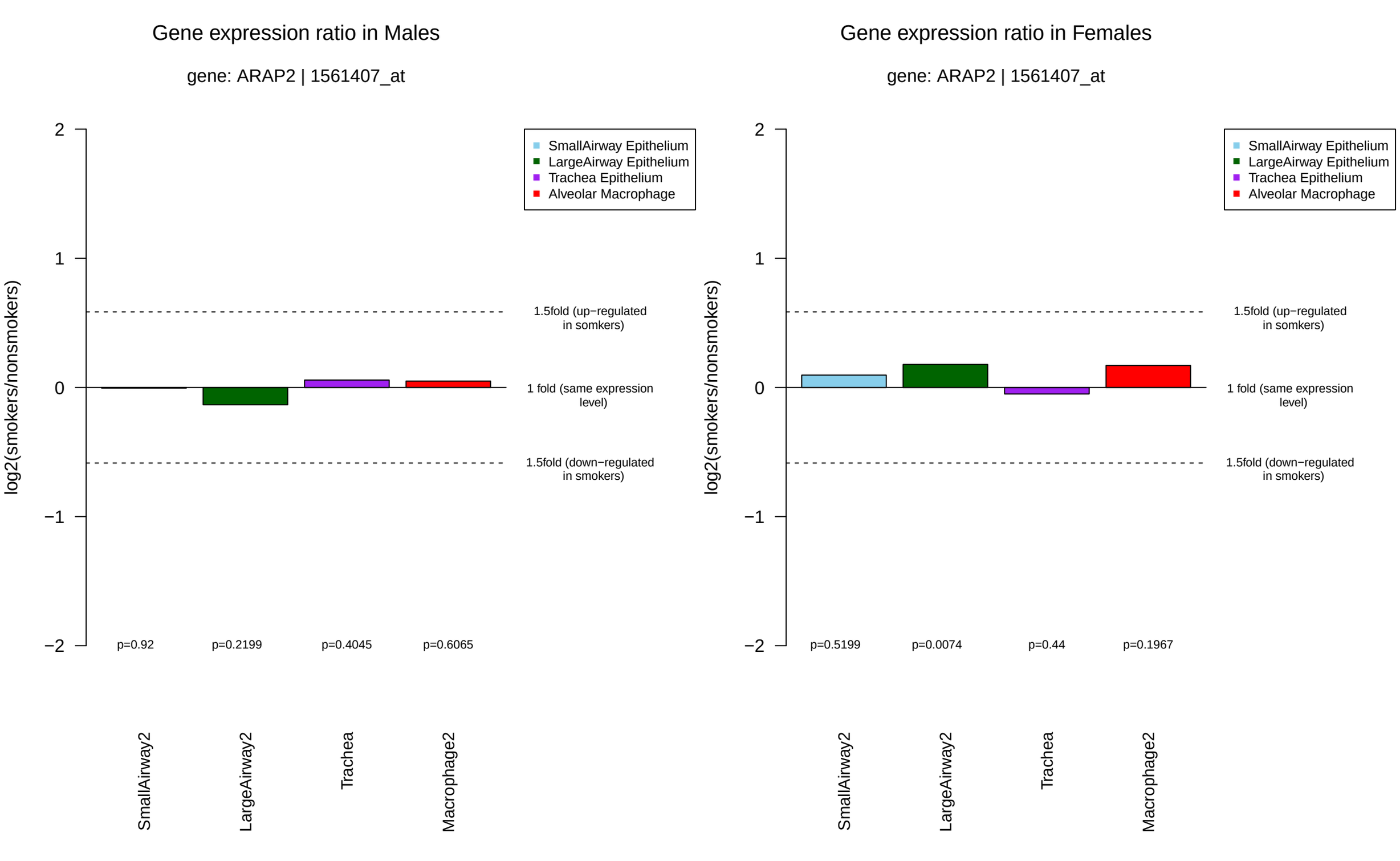
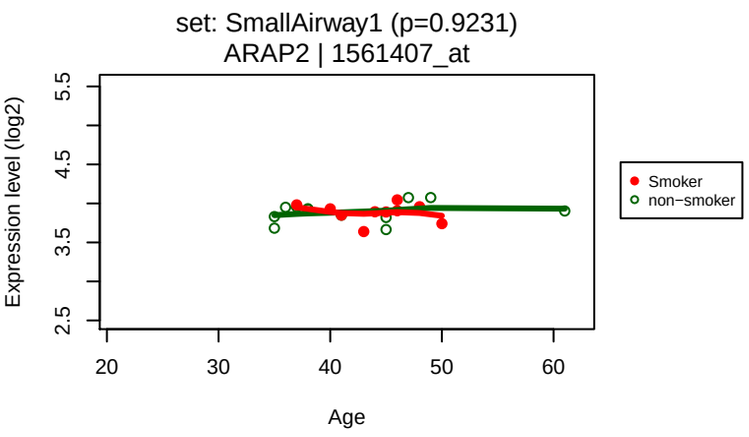
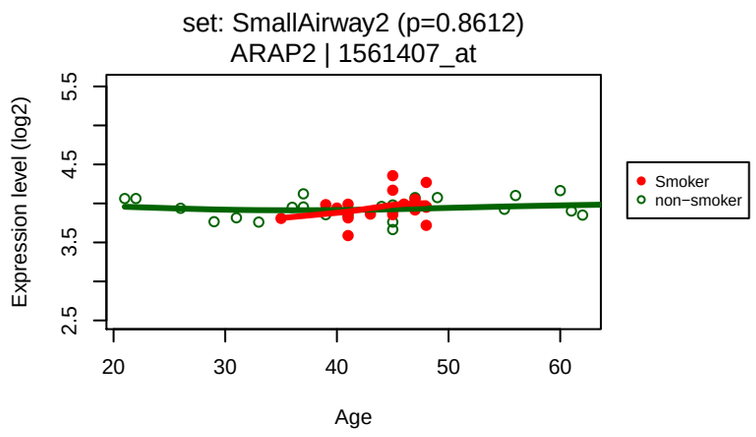
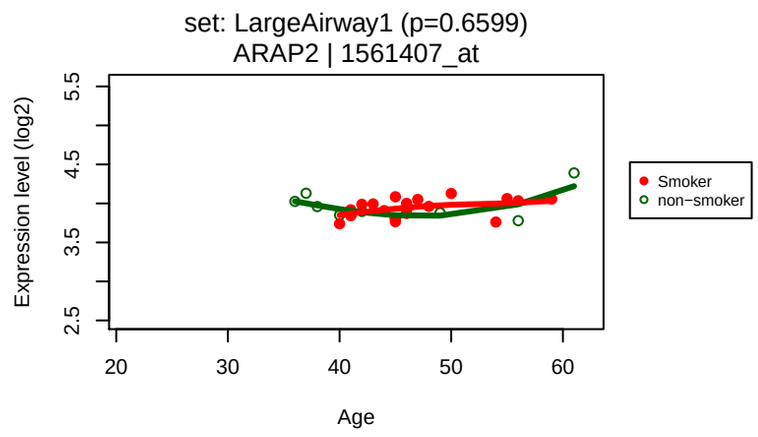

Gene expression of smokers and nonsmokers
gene: ARAP2 | 1561407_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9231
p=0.8612
p=0.7104
p=0.6599
p=0.6689
p=0.637
p=0.2368
p=0.211
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

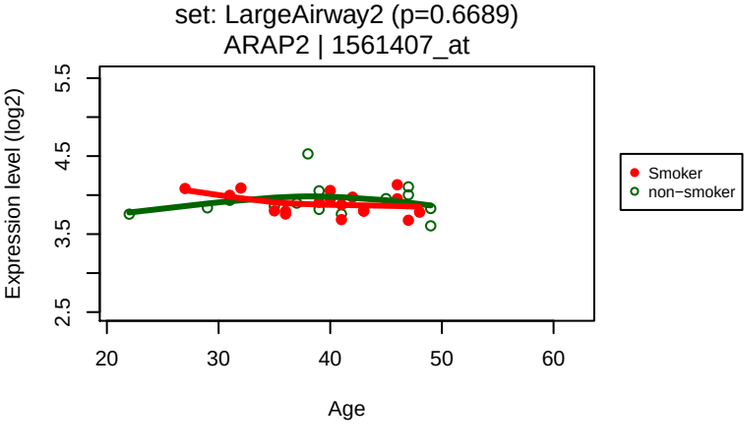

Gene expression ratio (smokers/nonsmokers)
gene: ARAP2 | 1561407_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9231
p=0.8612
p=0.7104
p=0.6599
p=0.6689
p=0.637
p=0.2368
p=0.211
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

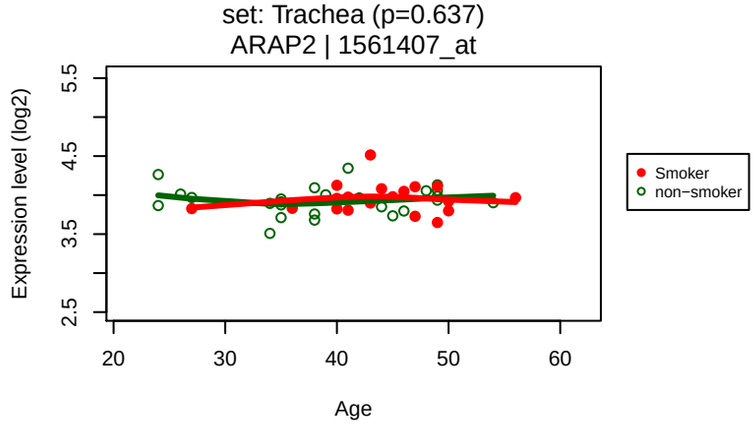

Gene expression ratio in Males
gene: ARAP2 | 1561407_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.92
p=0.2199
p=0.4045
p=0.6065
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARAP2 | 1561407_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5199
p=0.0074
p=0.44
p=0.1967
SmallAirway2
LargeAirway2
Trachea
Macrophage2

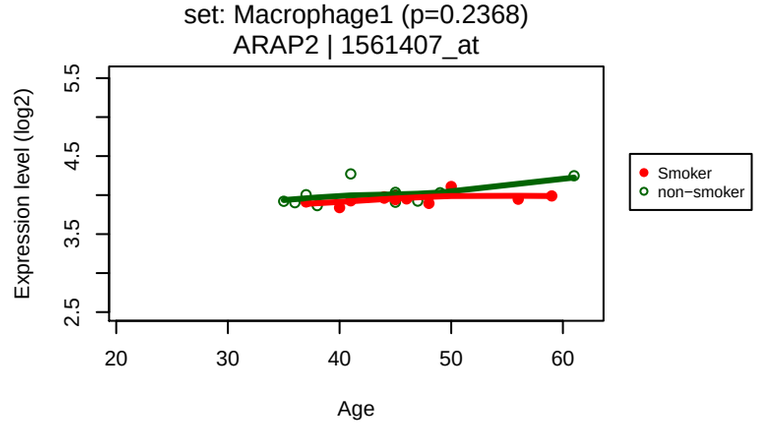

set: SmallAirway1 (p=0.9231)
ARAP2 | 1561407_at
Expression level (log2)
Age
Smoker
non−smoker

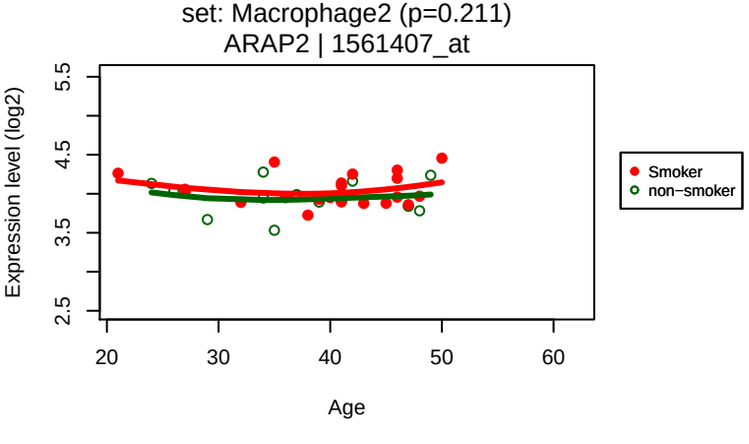

set: SmallAirway2 (p=0.8612)
ARAP2 | 1561407_at
Expression level (log2)
Age
Smoker
non−smoker

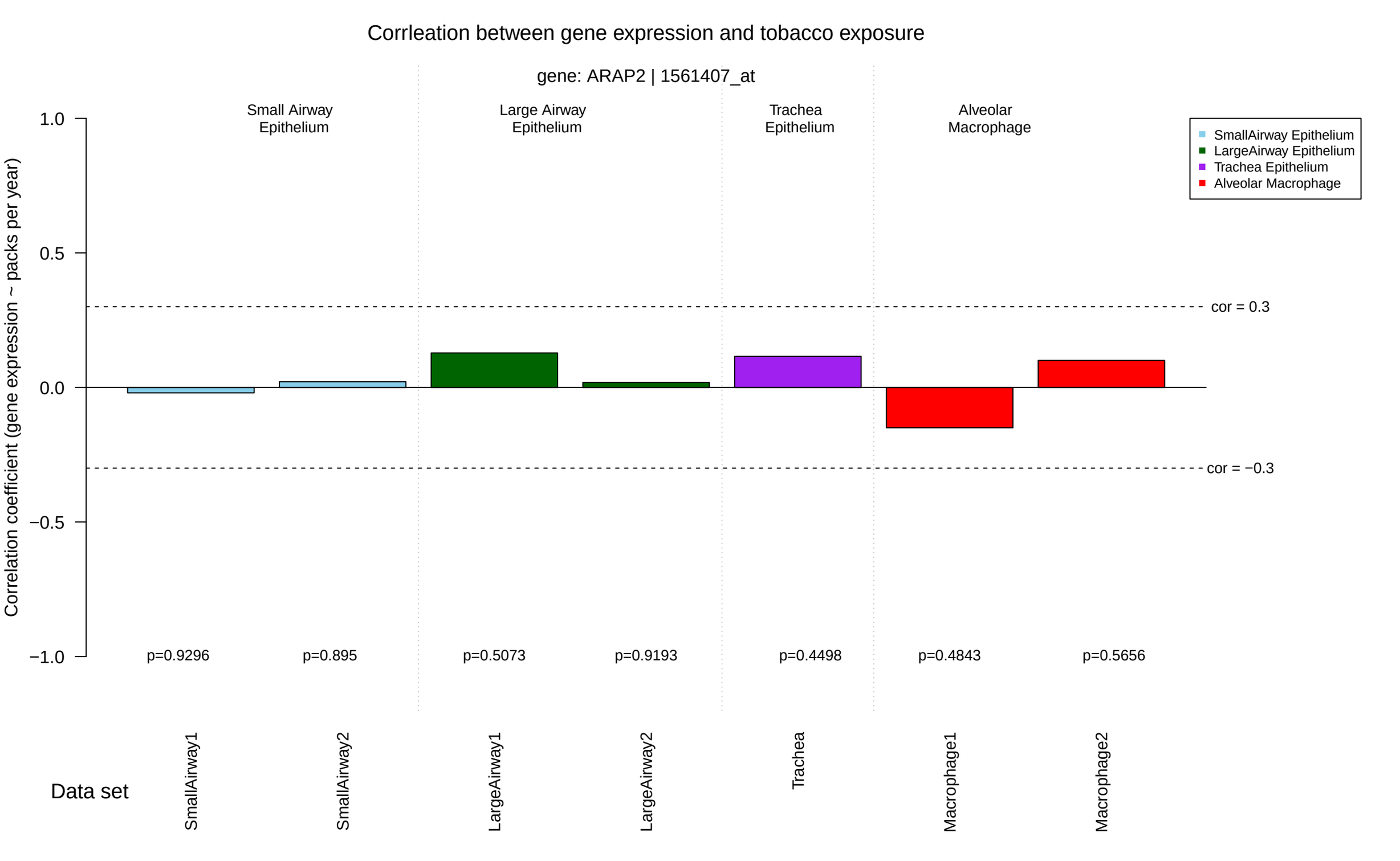

set: LargeAirway1 (p=0.6599)
ARAP2 | 1561407_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6689)
ARAP2 | 1561407_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.637)
ARAP2 | 1561407_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2368)
ARAP2 | 1561407_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.211)
ARAP2 | 1561407_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARAP2 | 1561407_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9296
p=0.895
p=0.5073
p=0.9193
p=0.4498
p=0.4843
p=0.5656
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2